FRC SENSOR CAN — REVISION B-2 PROTOTYPE

Three industrial / logic inputs. Protected battery and regulated 5 V sensor power.
STM32F042 + TCAN3413. Digital status and edge counts over a custom team CAN protocol.
Not a CTRE Phoenix-compatible device. Inspect and bench-test prototypes before robot use.

Battery entry and internal supplies

File: power.kicad_sch

Protected sensor power rails

File: sensor_power.kicad_sch

Controller, clock, debug and status

File: controller.kicad_sch

CAN physical interface and termination

File: can.kicad_sch

Input 1: industrial / logic selection

File: input1.kicad_sch

Input 2: industrial / logic selection

File: input2.kicad_sch

Input 3: industrial / logic selection

File: input3.kicad_sch

Fixed 5 V boost and reservoir

File: boost.kicad_sch

Original design — custom WPILib CAN protocol

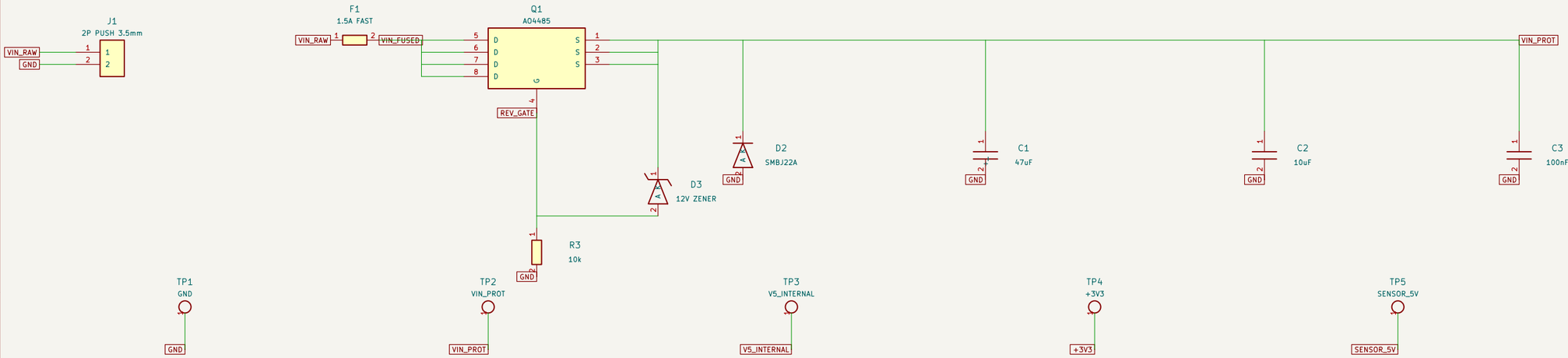
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File: FRC_Sensor_CAN.kicad_sch

Title: FRC Sensor CAN

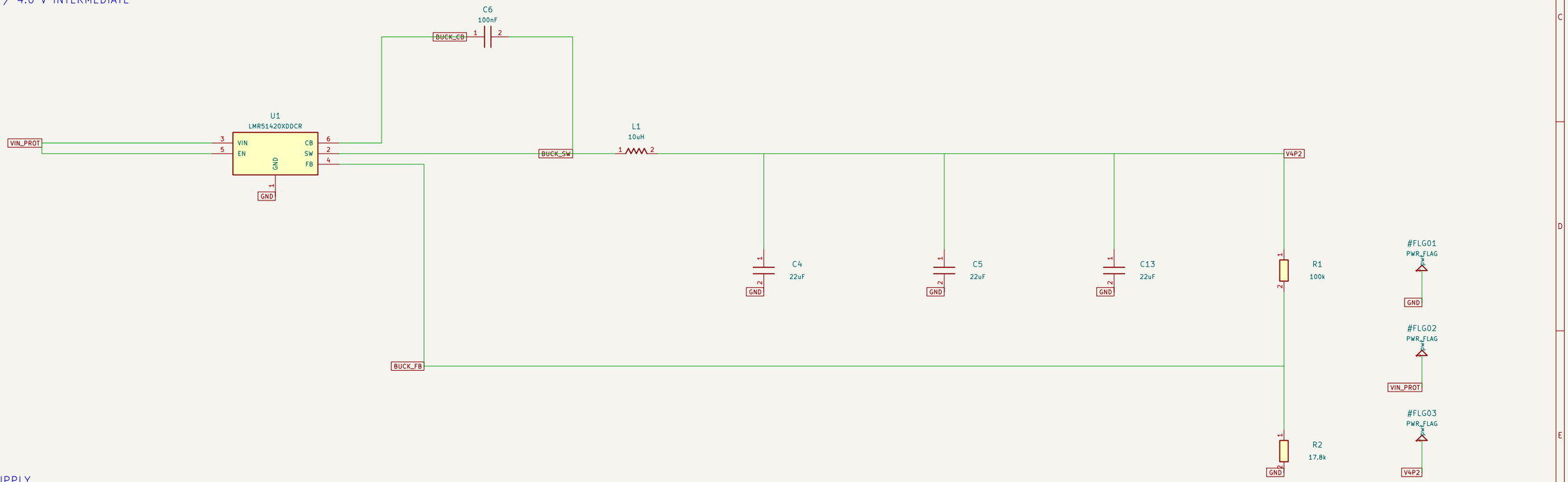
Size: A3	Date: 2026-09-08	**Rev: B-2 PROTOTYPE**
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Battery entry and internal supplies

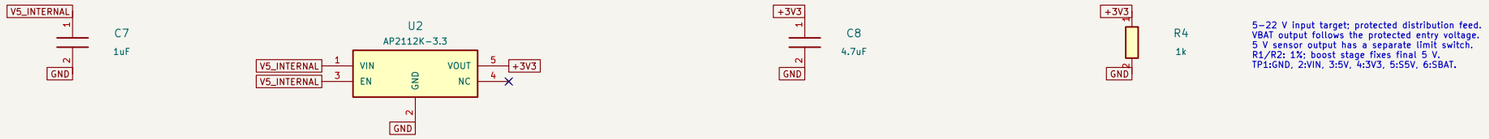
PROTECTED BATTERY ENTRY



500 kHz BUCK / 4.0 V INTERMEDIATE



3.3 V LOGIC SUPPLY



5–22 V input target; protected distribution feed.
VBAT output follows the protected entry voltage.
5 V sensor output has a separate limit switch.
R1/R2: 1%; boost stage fixes final 5 V.
TP1:GND, 2:VIN, 3:5V, 4:3V3, 5:S5V, 6:SBAT.

Prototype: bench validation required before robot use

Original sensor interface design

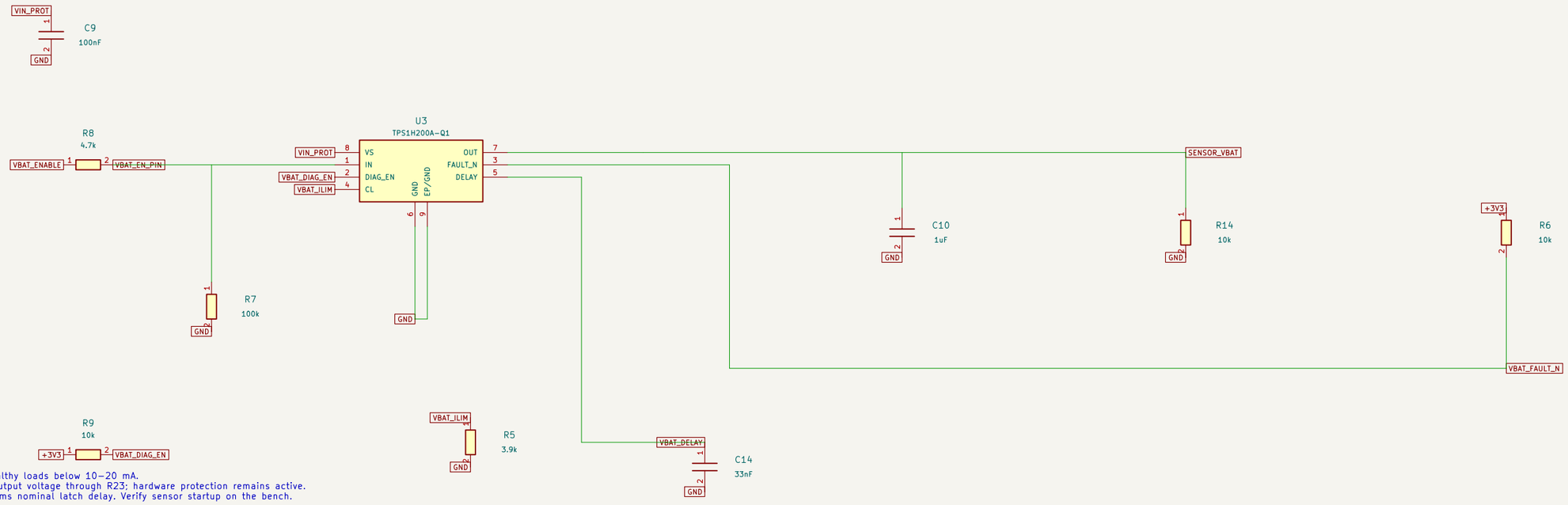
Sheet: /Battery entry and internal supplies/
File: power.kicad_sch

Title: FRC Sensor CAN — Battery entry and internal supplies

Size: A3	Date: 2026-09-08	**Rev: B-2 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 2/9

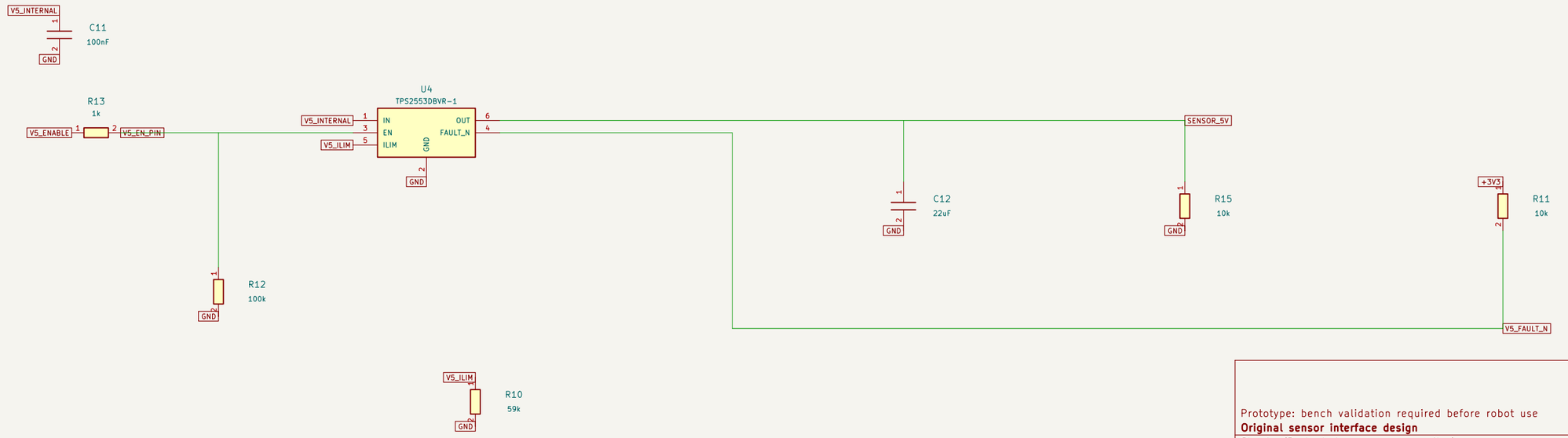
Protected sensor power rails

PROTECTED BATTERY PASSTHROUGH



U3 FAULT also asserts for healthy loads below 10–20 mA.
Firmware monitors switched output voltage through R23; hardware protection remains active.
C14 = 33 nF gives about 11 ms nominal latch delay. Verify sensor startup on the bench.

REGULATED 5 V SENSOR OUTPUT



Usable targets: 300 mA total on each rail, shared across all ports.
Both enable pins have hardware pulldowns. Firmware latches overload faults.
TPS2553 -1 is the latching version. Clear only after inspecting sensor wiring.

Prototype: bench validation required before robot use
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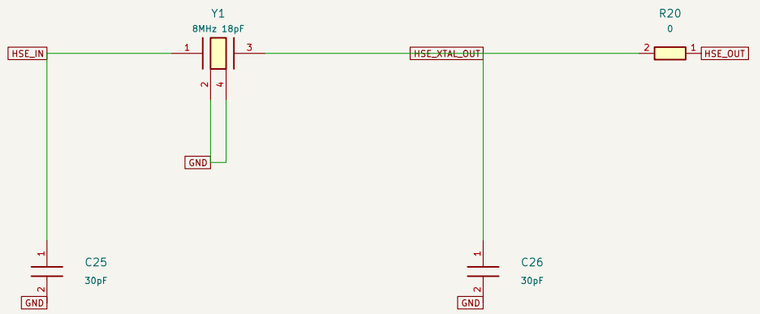
Sheet: /Protected sensor power rails/
File: sensor_power.kicad_sch

Title: FRC Sensor CAN — Protected sensor power rails

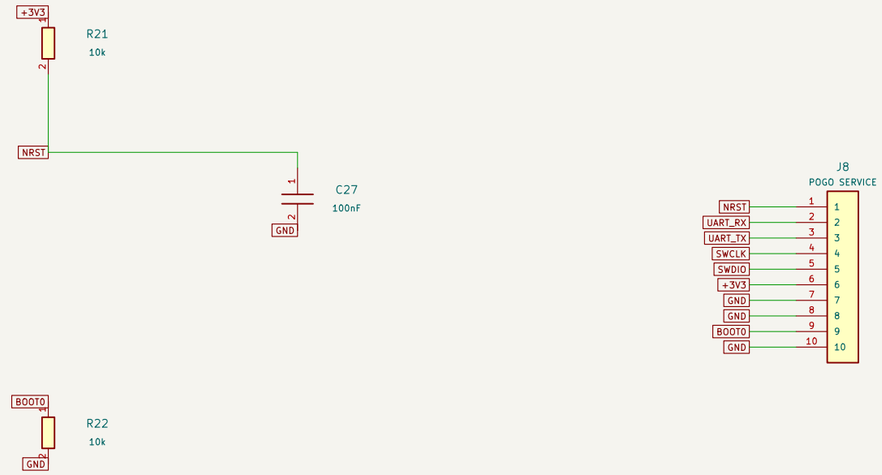
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KiCad E.D.A. 10.0.6		Id: 3/9

Controller, clock, debug and status

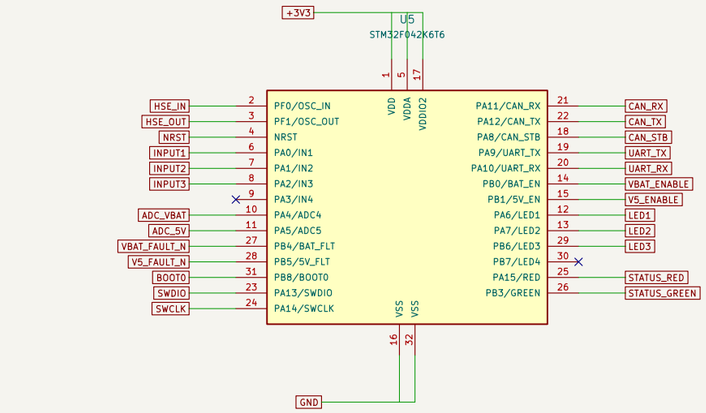
8 MHz CRYSTAL / PLL x6

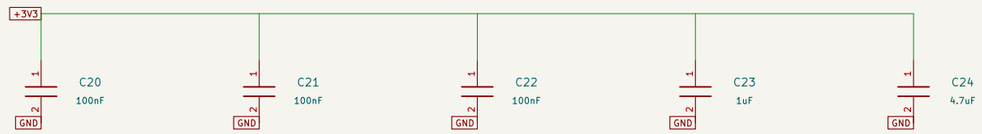


RESET / BOOT / SERVICE



LOCAL DECOUPLING





RAIL MONITORING



J8 is a TC2050-IDC-NL pogo footprint.
CUSTOM pinout: pin 6 is 3.3 V reference OUTPUT.
Use a 3.3 V SWD / UART adapter and shared GND.
BOOT0 normally low; reset is on J8 pin 1.
CAN PA11/PA12 use AF4. LEDs are on CAN sheet.

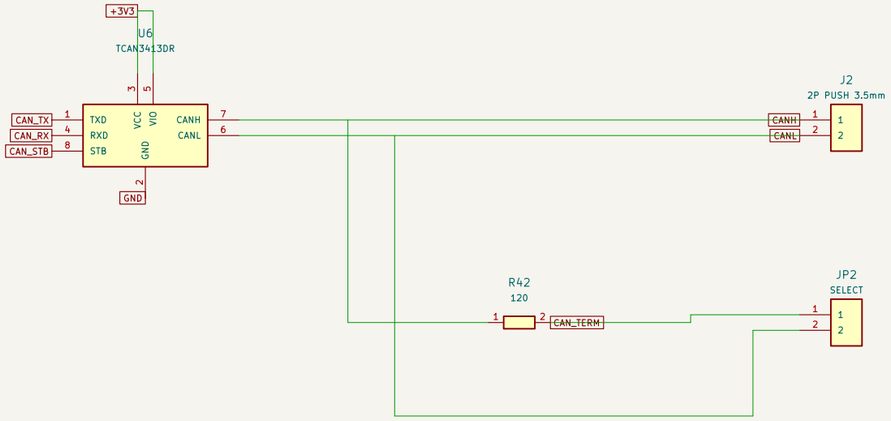
Prototype: bench validation required before robot use
Original sensor interface design

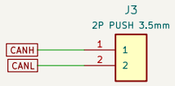
Sheet: /Controller, clock, debug and status/
File: controller.kicad_sch

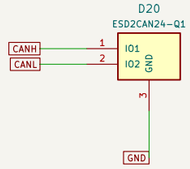
Title: FRC Sensor CAN — Controller, clock, debug and status

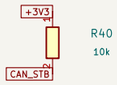
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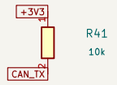
CAN physical interface and termination



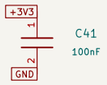


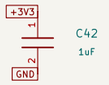






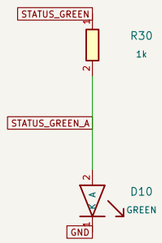


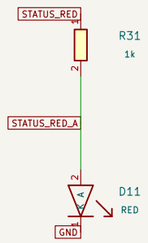




1 Mbit/s. Fit JP2 only at a physical CAN trunk end.
TCAN3413 uses 3.3 V for both VCC and VIO.
STB pull-up keeps the bus silent during MCU reset.
CAN ground is shared through the power connection.

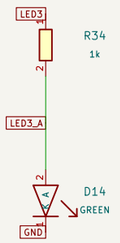
STATUS AND THREE INPUT INDICATORS











Prototype: bench validation required before robot use

Original sensor interface design

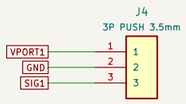
Sheet: /CAN physical interface and termination/
File: can.kicad_sch

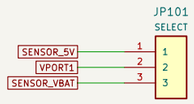
Title: FRC Sensor CAN — CAN physical interface and termination

Size: A3	Date: 2026-09-08	**Rev: B-2 PROTOTYPE**
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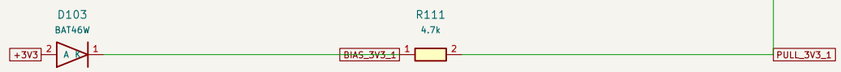
Input 1: industrial / logic selection

PORT POWER AND BIAS







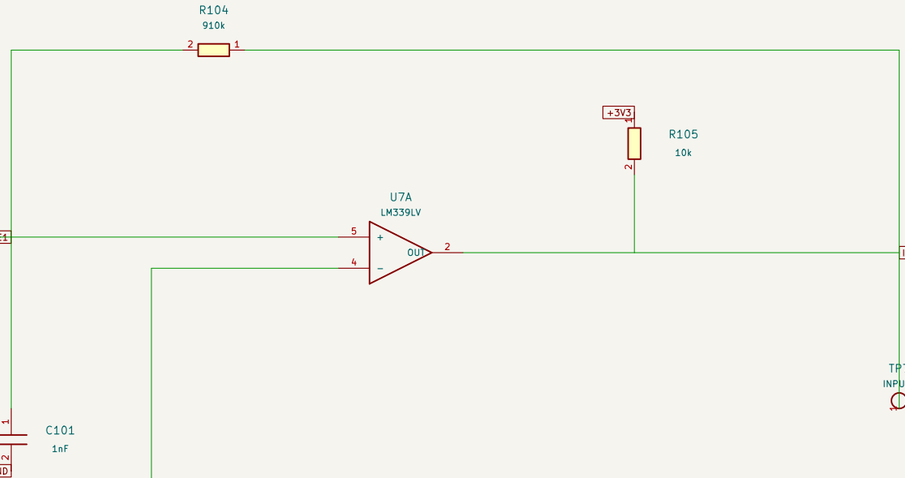


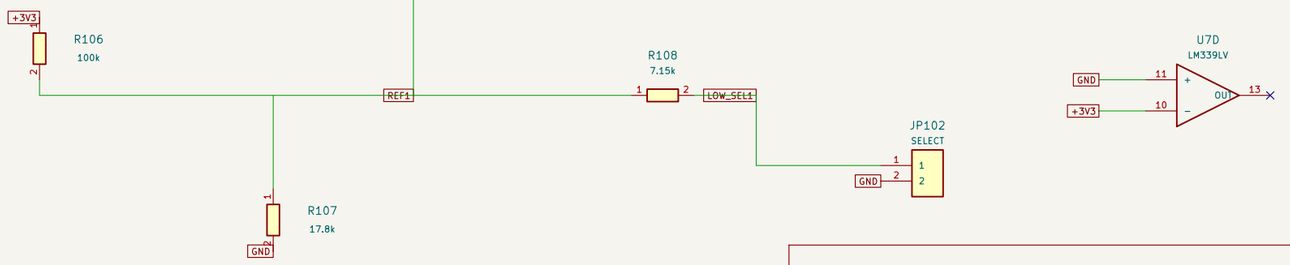


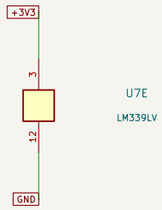


One shunt per selector. Bias diodes block signal backfeed into the supply rails.
NPN: port pullup. PNP: pulldown. Push–pull: bias open. Each bias branch is current limited.

PROTECTED SIGNAL AND HYSTERESIS









Threshold jumper open: LOW <=4V, HIGH >=7V.
Jumper fitted: LOW <=0.8V, HIGH >=2.4V.
0–30V signal target. 1nF input filter; digital debounce in firmware.
Do not apply external power to VPORT.

Prototype: bench validation required before robot use
Original sensor interface design

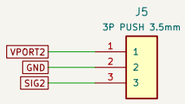
Sheet: /Input 1: industrial / logic selection/
File: input1.kicad_sch

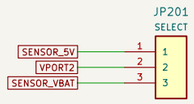
Title: FRC Sensor CAN – Input 1: industrial / logic selection

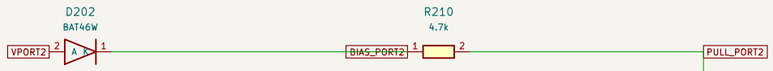
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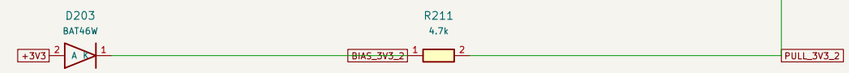
Input 2: industrial / logic selection

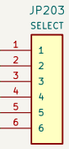
PORT POWER AND BIAS







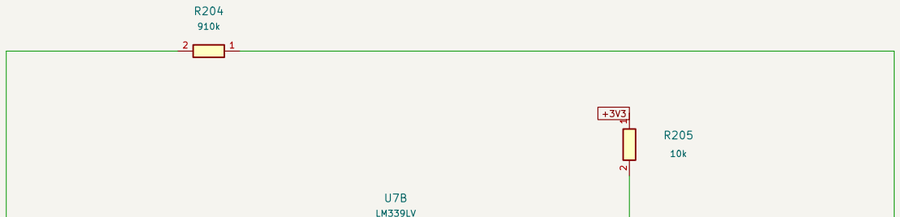


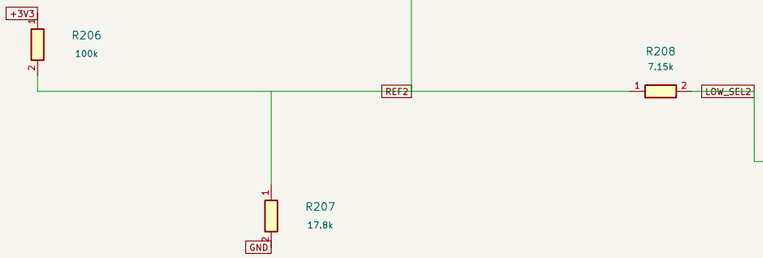




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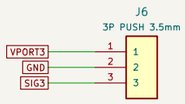
Sheet: /Input 2: industrial / logic selection/
File: input2.kicad_sch

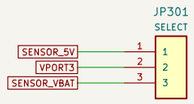
Title: FRC Sensor CAN – Input 2: industrial / logic selection

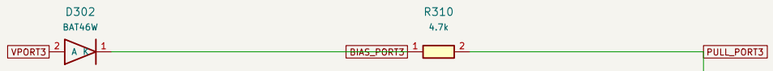
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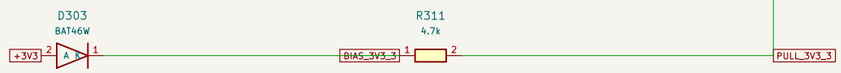
Input 3: industrial / logic selection

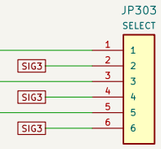
PORT POWER AND BIAS







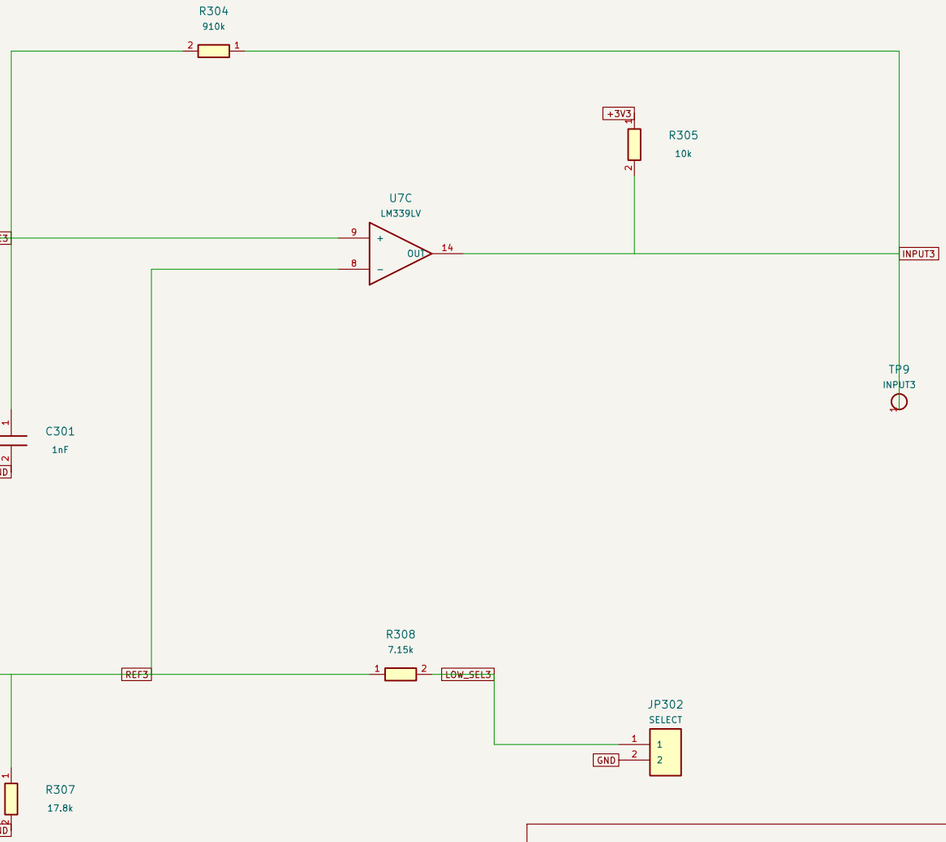






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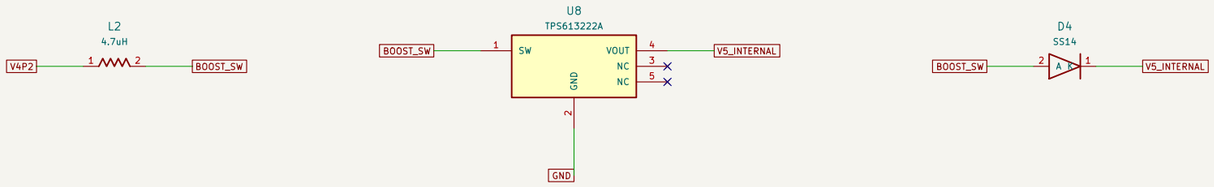
Sheet: /Input 3: industrial / logic selection/
File: input3.kicad_sch

Title: FRC Sensor CAN – Input 3: industrial / logic selection

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Fixed 5 V boost and reservoir

4.0 V INTERMEDIATE TO FIXED 5 V



TPS613222A: pins 1 SW, 2 GND, 4 OUT; pins 3/5 NC.
D4 external Schottky shares rectifier current.
C15/C16: two 22 uF 25 V ceramics.
Target: 300 mA usable sensor load plus board logic.
Qualify startup, ripple, thermal rise and cascaded-loop interaction at 5/12/22 V.

Prototype: bench validation required before robot use
Original sensor interface design

Sheet: /Fixed 5 V boost and reservoir/
File: boost.kicad_sch

Title: FRC Sensor CAN — Fixed 5 V boost and reservoir

Size: A3	Date: 2026-09-08	**Rev: B-2 PROTOTYPE**
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